

to impersonate legitimate users (masquerade), replay, retransmit or modify the content in transit, and launch denial-of-service (DoS) attacks (Fig. 1).

Every computer on a network operates on the seven-layer architecture of Open System Interconnection (OSI) model developed by International Telecommunication Union (ITU). Specific functions of each layer and possible cyber attacks are shown in Fig. 2. It is noteworthy here that the entire security system is only as secure as the single weakest layer.

ITU-T recommendation X.800—security architecture for OSI model—defines a systematic approach to assessing and providing security. However, in reality, security is often overlooked and use of security measures in networks is not growing as fast as the number and size of networks. That's why cyber attacks are becoming so common and easy.

Cyber threat tools are discussed below.

Viruses

A computer virus is a program that spreads from one computer to another by making copies of itself on a computer, or by inserting a piece of computer code into the program or operating system files. It has the capability to replicate itself. Virus attacks can be active or passive type because they can either damage files and/or affect a computer's performance and stability. A very common example of a virus infecting a computer or spreading is when you open an infected e-mail attachment.

Some e-mail viruses send confidential information in messages when spreading. Others do nothing beyond reproducing themselves. A computer virus may also send a Web address link as an instant message to all the contacts on an infected computer. As soon as this Web address link is accessed, the virus hosted at the link infects the new computer.

John Von Neumann, considered to be the theoretical father of computer virology, designed the world's first self-reproducing computer program

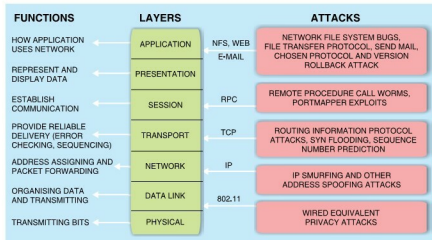


Fig. 2: OSI layers, their functions and possible attacks

in 1949. Before the proliferation of computer networks, most viruses were only able to spread themselves via removable media, particularly floppy disks. Traditional viruses emerged with the growth of computer networks. In a network system, viruses can increase their chances of replication and spreading to other systems by infecting files on the network file system that is accessed by other computers.

There is no single indicator of a virus infection, but some of the more common symptoms include poor computer performance, pop-up ads displaying even when a pop-up blocker is turned on or the Internet getting disconnected.

File-infecting viruses. File-infecting viruses are the most common viruses that attach themselves, i.e., inject code to executable files, to infect other programs and files. As soon as the user runs the infected file, the virus executes its own code to attach itself to other executable files on the user's computer. Obviously, the virus goes along for the ride when the user transfers infected files to another computer and infects more files onto the new computer.

Non-resident viruses. Non-resident viruses, when executed, immediately search for other computers to infect them and ultimately transfer control to the application program they infected. Such viruses consist of a finder module to search for new files and a replication module to infect these new files.

Resident viruses. Resident viruses consist of a replication module that they load into the memory on execution, and do not search for hosts when they are executed. These viruses ensure that the replication module is executed each time the operating system is called to perform a certain task. Upon execution, such viruses transfer control to the host program and remain active in the background to infect new hosts when those files are accessed by other application programs or the operating system itself.

Depending on the rate of infection-spread, resident viruses are roughly divided into two categories of fast and slow infectors. As the name suggests, slow infectors are designed deliberately to infect hosts infrequently so that detection of such viruses becomes very hard for anti-virus programs. Fast infectors, on the other hand, infect as many hosts (including anti-virus software program) as possible at a very fast pace. So these can become a 'piggy-back' on the anti-virus program itself and in this way infect all the files that are scanned. Detection of such viruses is easy because they heavily affect computer performance and perform several suspicious actions.

Cavity viruses. Cavity viruses take advantage of unused areas of portable executable files to overwrite these areas with their own small codes without increasing the size or damaging the executable files.